The documentation and process conversion measures necessary to comply with this document shall be completed by 5 September 2015.

INCH-POUND

MIL-PRF-19500/512L
5 June 2015
SUPERSEDING
MIL-PRF-19500/512K
w/AMENDMENT 1
29 January 2014

PERFORMANCE SPECIFICATION SHEET

* TRANSISTOR, PNP, SILICON, SWITCHING,
ENCAPSULATED (THROUGH-HOLE AND SURFACE MOUNT) AND UNENCAPSULATED,
RADIATION HARDNESS ASSURANCE, DEVICE TYPES 2N4029 AND 2N4033,
QUALITY LEVELS: JAN, JANTX, JANTXV, JANS, JANHC AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL–PRF–19500.

1. SCOPE

* 1.1 Scope. This specification covers the performance requirements for PNP silicon transistors for use in high speed switching and driver applications. Four levels of product assurance (JAN, JANTX, JANTXV, and JANS) are provided for each encapsulated device type as specified in MIL–PRF–19500, and two levels of product assurance (JANHC and JANKC) for each unencapsulated device type. Provisions for radiation hardness assurance (RHA), eight radiation levels is provided for JANTXV and JANS product assurance levels. RHA level designators “M”, “D”, “P“, “L”, “R”, “F’, “G”, and “H” are appended to the device prefix to identify devices, which have passed RHA requirements.

* 1.2 Package and die outlines. The device package for the encapsulated device types are as follows: TO-18 in accordance with figure 1, TO-39 in accordance with figure 2, surface mount UA in accordance with figure 3, surface mount UB in accordance with figure 4. The dimensions and topography for JANHC and JANKC unencapsulated die are in accordance with figures 5 and 6.

1.3 Maximum ratings. Unless otherwise specified T_A = +25°C.

V_{CBO}	V_{CEO}	V_{EBO}	I_C	T_J and T_{STG}
V dc	V dc	V dc	A dc	°C
80	80	5.0	1.0	-65 to +200

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil . Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil .

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FSC 5961

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1.3 Maximum ratings - continued.

Types	P_T T_A = +25°C (1) (2)	P_T T_C = +25°C (1) (2)	P_T $T_{SP(IS)}$ = +25°C (1) (2)	$R_{\theta JA}$ (2) (3)	$R_{\theta JC}$ (2) (3)	$R_{\theta JSP(IS)}$ (2) (3)	$R_{\theta JSP(AM)}$ (2) (3)
	W	W	W	°C/W	°C/W	°C/W	°C/W
2N4033	0.800	4	N/A	195	40	N/A	N/A
2N4029	0.500	1	N/A	325	150	N/A	N/A
2N4033UA	0.500	N/A	1.5	325	N/A	110	40
2N4033UB	(4) 0.500	N/A	1.5	325	N/A	90	N/A

(1) For derating, see figures 7, 8, 9, 10, and 11.
(2) See 3.3.
(3) For thermal impedance curves, see figures 12, 13, 14, 15, 16, 17, and 18.
(4) For non-thermal conductive PCB or unknown PCB surface mount conditions in free air, substitute figures 8 and 16 for the UB package and use $R_{\theta JA}$.

1.4 Primary electrical characteristics. Unless otherwise specified T_A = +25°C.

Limits	h_{FE1} V_{CE} = 5.0 V dc I_C = 100 µA dc	h_{FE2} V_{CE} = 5.0 V dc I_C = 100 mA dc	h_{FE3} V_{CE} = 5.0 V dc I_C = 500 mA dc	h_{FE4} V_{CE} = 5.0 V dc I_C = 1.0 A dc	$\lvert h_{fe} \rvert$ f = 100 MHz V_{CE} = 10 V dc I_C = 50 mA dc
Min	50	100	70	25	1.5
Max		300			6.0

Limits	$V_{CE(SAT)2}$ I_C = 500 mA dc I_B = 50 mA dc	C_{obo} V_{CB} = 10 V dc I_E = 0 100 kHz ≤ f ≤ 1 MHz	t_d	t_r	t_s	t_f
	V dc	pF	ns	ns	ns	ns
Min						
Max	0.5	20	15	25	175	35

* 1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL–PRF–19500, and as specified herein. See 6.5 for PIN construction example and 6.6 for a list of available PINs.

* 1.5.1 JAN brand and quality level designators.

* 1.5.1.1 Encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: "JAN", "JANTX", "JANTXV", and "JANS".

* 1.5.1.2 Unencapsulated devices (die). The quality level designators for unencapsulated devices (die) that are applicable for this specification sheet from the lowest to the highest level are as follows: "JANHC" and "JANKC".

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* 1.5.2 Radiation hardness assurance (RHA) designator. The RHA levels that are applicable for this specification sheet from lowest to highest are as follows: "M", "D", "P", "L", "R", "F", "G", and "H").

* 1.5.3 Device type. The designation system for the device types of transistors covered by this specification sheet are as follows.

* 1.5.3.1 First number and first letter symbols. The transistors of this specification sheet use the first number and letter symbols "2N".

* 1.5.3.2 Second number symbols. The second number symbols for the transistors covered by this specification sheet are "4029".

* 1.5.4 Suffix symbols. The following suffix symbols are incorporated in the PIN as applicable.

* 1.5.4.1 First suffix symbol. The first suffix symbol "A" indicates that the transistor is a modified version of the approved device type.

* 1.5.4.2 Following suffix symbols. The following suffix symbols are incorporated in the PIN for this specification sheet:

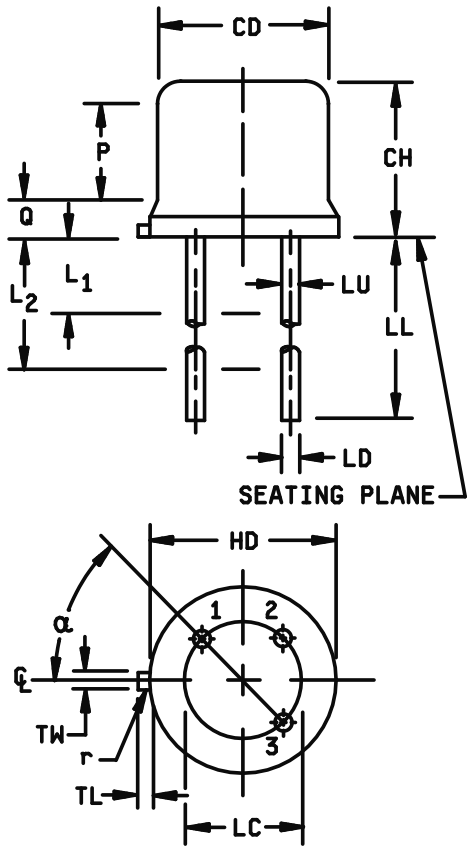
	A blank second suffix symbol indicates a through-hole mount package (see figure 1 and figure 2).
UA	Indicates a 4 pad surface mount package (see figure 3).
UB	Indicates a 4 pad surface mount package. The metal lid is connected to pad 4 (see figure 4).

* 1.5.5 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS–19500.

* 1.5.6 Die identifiers for unencapsulated devices (manufacturers and critical interface identifiers). The manufacturer die identifiers that are applicable for this specification sheet is "A" (see figure 5) and “B” (see figure 6) and 6.5.

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CD	.178	.195	4.52	4.95	
CH	.170	.210	4.32	5.33	
HD	.209	.230	5.31	5.84	
LC	.100 TP		2.54 TP		6
LD	.016	.021	0.41	0.53	7, 8
LL	.500	.750	12.70	19.05	7, 8, 12
LU	.016	.019	0.41	0.48	7, 8
L_1		.050		1.27	7. 8
L_2	.250		6.35		7, 8
Q		.040		1.02	5
TL	.028	.048	0.71	1.22	3, 4
TW	.036	.046	0.91	1.17	3
R		.010		0.25	10
P	.100		2.54		
α	45°TP		45°TP		6



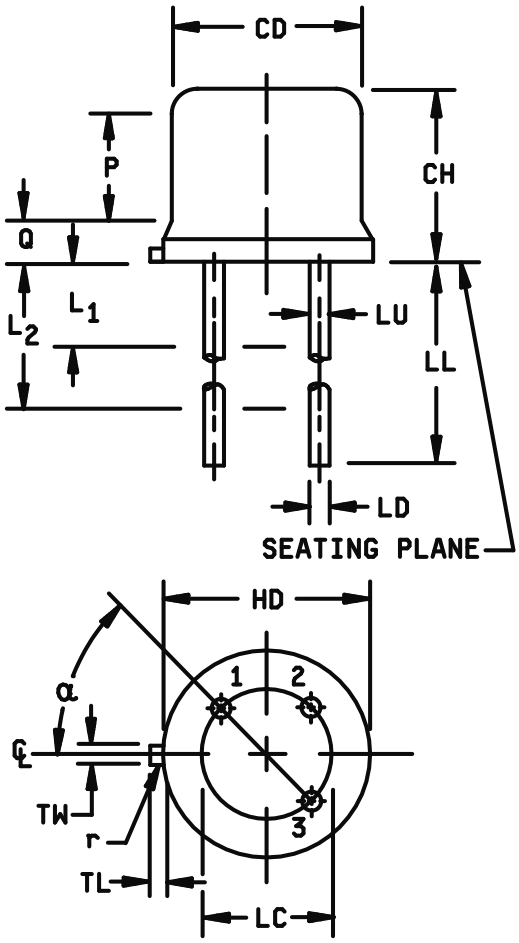
NOTES:

1. Dimensions are in inches.
2. Millimeters equivalents are given for general information only.
3. Beyond r (radius) maximum, TW shall be held for a minimum length of .011 (0.28 mm).
4. Dimension TL measured from maximum HD.
5. Body contour optional within zone defined by HD, CD, and Q.
6. Leads at gauge plane .054 +.001 -.000 inch (1.37 +0.03 –0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC. The device may be measured by direct methods.
7. Dimension LU applies between L_1 and L_2. Dimension LD applies between L_2 and minimum. Diameter is uncontrolled in L_1 and beyond LL minimum.
8. All three leads.
9. The collector shall be internally connected to the case.
10. Dimension r (radius) applies to both inside corners of tab.
11. In accordance with ASME Y14.5M, diameters are equivalent to θx symbology.
12. For “L” suffix devices, dimension LL is 1.50 (38.10 mm) minimum, 1.75 (44.45 mm) maximum.

FIGURE 1. Physical dimensions (type 2N4029) (TO-18).

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Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
CD	.305	.335	7.75	8.51	
CH	.240	.260	6.10	6.60	
HD	.335	.370	8.51	9.40	
LC	.200 TP		5.08 TP		6
LD	.016	.021	0.41	0.53	7, 8
LL	.500	.750	12.70	19.05	7, 8, 12
LU	.016	.019	0.41	0.48	7, 8
L_1		.050		1.27	7, 8
L_2	.250		6.35		7, 8
Q		.050		1.27	5
TL	.029	.045	0.74	1.14	3, 4
TW	.028	.034	0.71	0.86	3
R		.010		0.25	10
P	.100		2.54		
α	45°TP		45°TP		6

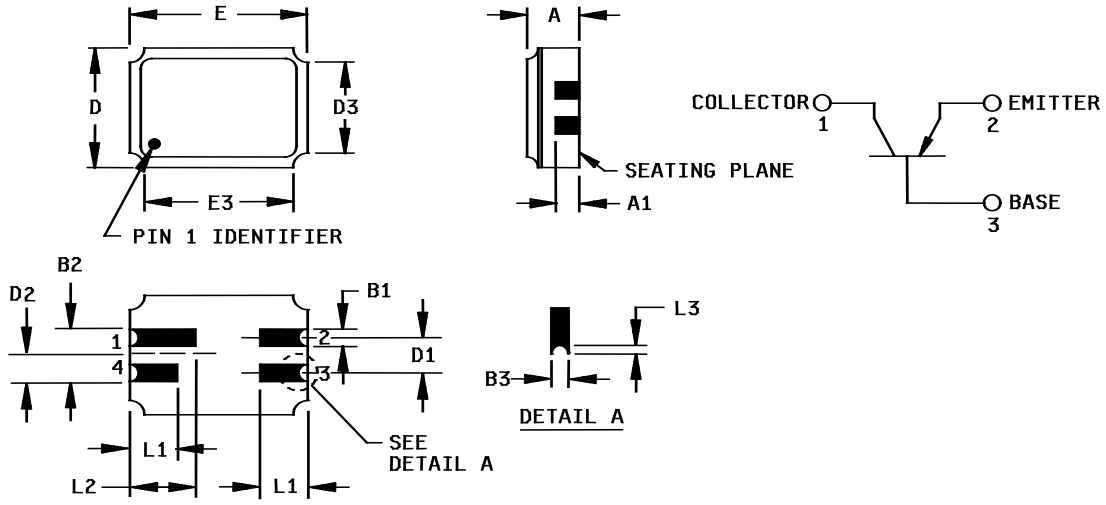


NOTES:

1. Dimensions are in inches.
2. Millimeters equivalents are given for general information only.
3. Beyond r (radius) maximum, TW shall be held for a minimum length of .011 (0.28 mm).
4. Dimension TL measured from maximum HD.
5. Body contour optional within zone defined by HD, CD, and Q.
6. Leads at gauge plane .054 +.001 -.000 inch (1.37 +0.03 -0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC. The device may be measured by direct methods.
7. Dimension LU applies between L_1 and L_2. Dimension LD applies between L_2 and minimum. Diameter is uncontrolled in L_1 and beyond LL minimum.
8. All three leads.
9. The collector shall be internally connected to the case.
10. Dimension r (radius) applies to both inside corners of tab.
11. In accordance with ASME Y14.5M, diameters are equivalent to θx symbology.
12. For “L” suffix devices, dimension LL is 1.50 (38.10 mm) minimum, 1.75 (44.45 mm) maximum.

FIGURE 2. Physical dimensions (type 2N4033) (TO-39).

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Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A	.061	.075	1.55	1.91	3
A_1	.029	.041	0.74	1.04	
B_1	.022	.028	0.56	0.71	
B_2	.075 REF		1.91 REF		
B_3	.006	.022	0.15	0.56	5
D	.145	.155	3.68	3.9	
D_1	.045	.055	1.14	1.39	

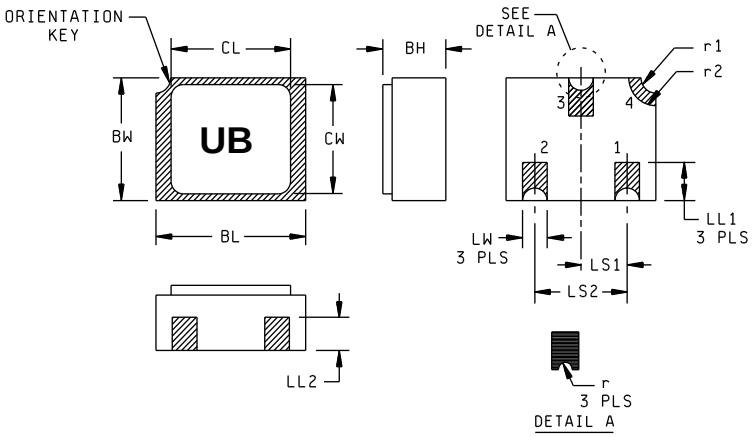
Ltr	Dimensions				Notes
	Inches		Millimeter		
	Min	Max	Min	Max	
D_2	.0375 BSC		0.952 BSC		
D_3		.155		3.94	
E	.215	.225	5.46	5.72	
E_3		.225		5.72	
L_1	.032	.048	0.81	1.22	
L_2	.072	.088	1.83	2.24	
L_3	.003		0.08		5

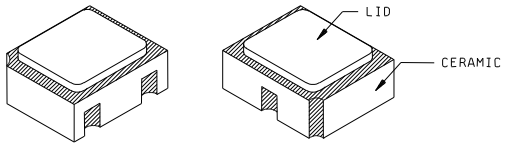
NOTES:

1. Dimensions are in inches.
2. Millimeters equivalents are given for general information only.
3. Dimension “A” controls the overall package thickness. When a window lid is used, dimension “A” must increase by a minimum of .010 inch (0.254 mm) and a maximum of .040 inch (1.020 mm).
4. The corner shape (square, notch, radius, etc.) may vary at the manufacturer’s option, from that shown on the drawing.
5. Dimensions “B3” minimum and “L3” minimum and the appropriately castellation length define an unobstructed three-dimensional space traversing all of the ceramic layers in which a castellation was designed. (Castellations are required on bottom two layers, optional on top ceramic layer.) Dimension “B3” maximum define the maximum width and depth of the castellation at any point on its surface. Measurement of these dimensions may be made prior to solder dipping.

FIGURE 3. Physical dimensions, surface mount (UA version).

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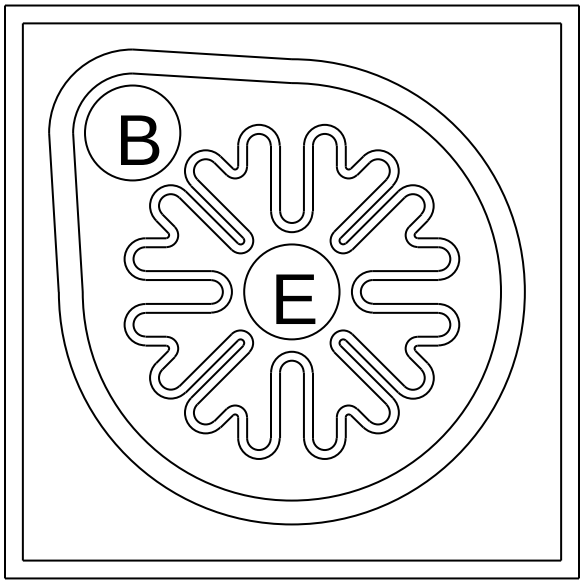
Symbol	Dimensions				Note
	Inches		Millimeters		
	Min	Max	Min	Max	
BH	.046	.056	1.17	1.42	
BL	.115	.128	2.92	3.25	
BW	.085	.108	2.16	2.74	
CL		.128		3.25	
CW		.108		2.74	
LL1	.022	.038	0.56	0.97	
LL2	.017	.035	0.43	0.89	

Symbol	Dimensions				Note
	Inches		Millimeters		
	Min	Max	Min	Max	
LS_1	.036	.040	0.91	1.02	
LS_2	.071	.079	1.80	2.01	
LW	.016	.024	0.41	0.61	
r		.008		.203	
r1		.012		.305	
r2		.022		.559	

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Hatched areas on package denote metallized areas
4. Pad 1 = Base, Pad 2 = Emitter, Pad 3 = Collector, Pad 4 = Shielding connected to the lid.
5. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 4. Physical dimensions, surface mount UB version.

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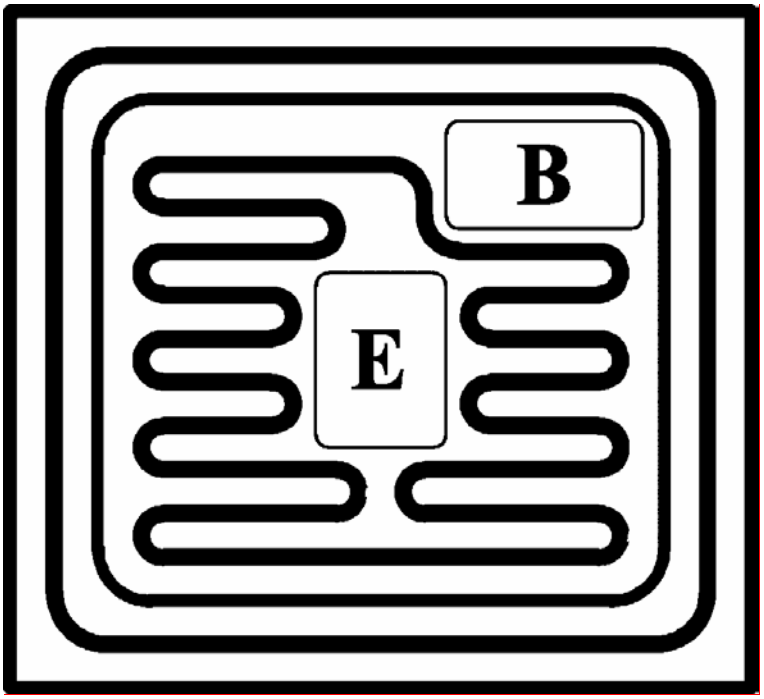


NOTES:

1. Die size: .030 x .030 inch (0.762 x 0.762 mm).
2. Die thickness: .008 ±.0016 inch (0.2032 ±0.04064 mm).
3. Base pad: .005 inch diameter (0.127 mm).
4. Emitter pad: .005 inch diameter (0.127 mm).
5. Back metal: Gold, 6,500 ±1,950 Å.
6. Top metal: Aluminum, 22,500 ±2,500 Å.
7. Back side: Collector.
8. Glassivation: SiO_2, 7,500 ±1,500 Å.

FIGURE 5. JANHC and JANKC (A-version) die dimensions.

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NOTES:

1.	Die size	.024 x .026 inch ±.002 inch (0.61 x 0.66 millimeters).
2.	Die thickness	.010 inch ±.0015 inch nominal (0.254 x 0.038 millimeters).
3.	Top metal	Aluminum 15,000 Å minimum, 18,000 Å nominal.
4.	Back metal	Gold 3,500 Å minimum, 5,000 Å nominal.
5.	Backside	Collector.
6.	Bonding pad	B = .004 x .006 inch (0.102 x 0.152 millimeters). E = .004 x .0055 inch (0.102 x 0.140 millimeters).

FIGURE 6. JANHC and JANKC (B-version) die dimensions.

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2. APPLICABLE DOCUMENTS

* 2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL–PRF–19500 – Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL–STD–750 – Test Methods for Semiconductor Devices.

(Copies of these documents are available online at http://quicksearch.dla.mil.)

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL–PRF–19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL–PRF–19500 and as follows.

PCB ..... Printed circuit board.
$R_{\theta JA}$ ..... Thermal resistance junction to ambient.
$R_{\theta JC}$ ..... Thermal resistance junction to case.
$R_{\theta JSP(IS)}$..... Thermal resistance junction to solder pads (infinite sink mount to PCB).
$T_{SP(AM)}$ ..... Temperature of solder pads (adhesive mount to PCB).
$T_{SP(IS)}$..... Temperature of solder pads (infinite sink mount to PCB).
UA, UB..... Surface mount case outlines.

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* 3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL–PRF–19500, and herein. The device package styles shall be as follows: Three pin metal can (TO-18) in accordance with figure 1, three pin metal can (TO-39) in accordance with figure 2, four pad surface mount case outline UA in accordance with figure 3, four pad surface mount case outline UB in accordance with figure 4, and unencapsulated die in accordance with figure 5 and 6 for device types, JANHC and JANKC.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL–PRF–19500, MIL–STD–750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

* 3.5 Radiation hardness assurance (RHA). Radiation hardness assurance requirements and test levels shall be as defined in MIL–PRF–19500.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.7 Marking.

* 3.7.1 Through hole mount packages. Marking shall be in accordance with figures 1 and 2.

* 3.7.2 Surface mount packages. Marking shall be in accordance with figures 3 and 4. That marking on the UA and UB packages shall consist of an abbreviated part number, the date code, and the manufacturer's symbol or logo. The prefixes JAN, JANTX, JANTXV and JANS can be abbreviated as J, JX, JV, and JS respectively. The "2N" prefix and the "UB" and "UA" suffix can also be omitted. The radiation hardened designator shall immediately precede (or replace) the device "2N" identifier (depending upon the degree of abbreviation required).

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4, table I and table II).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL–PRF–19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table III tests, the tests specified in table III herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

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* 4.3 Screening

* 4.3.1 Screening of packaged devices (quality levels JANS, JANTX, and JANTXV only). Screening of packaged devices shall be in accordance with table E-IV of MIL-PRF-19500 and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen	Measurement	
	JANS level	JANTX and JANTXV levels
(1) (2) 3c	Thermal impedance method 3131 of MIL-STD-750 (see 4.3.1.2).	Thermal impedance method 3131 of MIL-STD-750 (see 4.3.1.2).
9	I_{CBO2} and h_{FE2}	Not applicable
11	I_{CBO2} and h_{FE2} ΔI_{CBO2} = 100 percent or 5 nA, whichever is greater; Δh_{FE2} = ±15 percent change from initial value.	I_{CBO2} and h_{FE2}
12	See 4.3.1.1	See 4.3.1.1
13	Subgroups 2 and 3 of table I herein; ΔI_{CBO2} = +100 percent of initial value or 5 nA, whichever is greater. Δh_{FE2} = ±15 percent change from initial value.	Subgroup 2 of table I herein; ΔI_{CBO2} = +100 percent of initial value or 5 nA, whichever is greater. Δh_{FE2} = ±15 percent change from initial value.

(1) Thermal impedance limits ($Z_{\theta JX}$) shall not exceed figures 12, 13, 14, 15, 16, 17, and 18.
(2) Shall be performed anytime after temperature cycling, screen 3a; JANTX and JANTXV levels do not need to be repeated in screening requirements.

4.3.1.1 Power burn-in conditions. Power burn-in conditions are as follows: V_{CB} = 10 - 30 V dc. Power shall be applied to achieve T_J = +135°C minimum using a minimum P_D = 75 percent of P_T maximum rated as defined in 1.3. With approval of the qualifying activity and preparing activity, alternate burn-in criteria (hours, bias conditions, T_J, and mounting conditions) may be used for JANTX and JANTXV quality levels. A justification demonstrating equivalence is required. In addition, the manufacturing site's burn-in data and performance history will be essential criteria for burn-in modification approval. Use method 3100 of MIL-STD-750 to measure T_J.

4.3.1.2 Thermal impedance. The thermal impedance measurements shall be performed on each die in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{MD} (and V_C where appropriate). Measured delay time (t_{MD}) = 70 μs maximum. See table III, subgroup 4 herein.

* 4.3.2 Screening of unencapsulated die (JANHC and JANKC). Screening of JANHC and JANKC unencapsulated die shall be in accordance with appendix G of MIL–PRF–19500. The burn-n duration of the JANKC level shall follow the JANS requirements, the JANHC levels shall follow the JANTX requirements of table E-IV of MIL–PRF–19500.

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* 4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL–PRF–19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL–PRF–19500, and table I.

* 4.4.2 Group B inspection.

* 4.4.2.1 Quality level JANS, table E-Via of MIL-PRF-19500. Group B inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table E-VIA (JANS) of MIL–PRF–19500. Delta requirements only apply to subgroups B4 and B5.

Subgroup	Method	Condition
B4	1037	V_{CB} = 10 V dc, 2,000 cycles, adjust device current, or power, to achieve a minimum ΔT_J of +100°C.
B5	1027	V_{CB} = 10 V dc; $P_D \geq$ 100 percent of maximum rated P_T (see 1.3). (NOTE: If a failure occurs, resubmission shall be at the test conditions of the original sample.) Option 1: 96 hours minimum sample size in accordance with MIL–PRF–19500, table E-VIa, adjust T_A or P_D to achieve T_J = +275°C minimum. Option 2: 216 hours minimum, sample size = 45, c = 0; adjust T_A or P_D to achieve a T_J = +225°C minimum.

* 4.4.2.2 Quality level (JAN, JANTX, and JANTXV), table E-VIb of MIL–PRF–19500. Separate samples may be used for each step. In the event of a lot failure, the resubmission requirements of MIL–PRF–19500 shall apply. In addition, all catastrophic failures during CI shall be analyzed to the extent possible to identify root cause and corrective action. Whenever a failure is identified as wafer lot or wafer processing related, the entire wafer lot and related devices assembled from the wafer lot shall be rejected unless an appropriate determined corrective action to eliminate the failure~~s~~ mode has been implemented and the devices from the wafer lot are screened to eliminate the failure mode. Delta requirements for JAN, JANTX, and JANTXV shall be after each step.

Step	Method	Condition
1	1026	Steady-state life: 1,000 hours minimum, V_{CB} = 10 V dc, power shall be applied to achieve T_J = +150°C minimum using a minimum of P_D = 75 percent of maximum rated P_T as defined in 1.3. n = 45 devices, c = 0. The sample size may be increased and the test time decreased as long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours.
2	1048	Blocking life, T_A = +150°C, V_{CB} = 80 percent of rated voltage, 48 hours minimum. n = 45 devices, c = 0.
3	1032	High-temperature life (non-operating), t = 340 hours, T_A = +200°C. n = 22, c = 0.

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4.4.2.3 Group B sample selection. Samples selected from group B inspection shall meet all of the following requirements:

a. For JAN, JANTX, and JANTXV, samples shall be selected randomly from a minimum of three wafers (or from each wafer in the lot) from each wafer lot. For JANS, samples shall be selected from each inspection lot. See MIL–PRF–19500.

b. Shall be chosen from an inspection lot that has been submitted to and passed table I, subgroup 2, conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (subgroups B4 and B5 for JANS, and group B for JAN, JANTX, and JANTXV) may be pulled prior to the application of final lead finish.

* 4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table E-VII of MIL–PRF–19500, and in 4.4.3.1 (JANS) and 4.4.3.2 (JAN, JANTX, and JANTXV) herein for group C testing. Endpoints shall be after each step and shall be in accordance with MIL–PRF–19500.

* 4.4.3.1 Quality level JANS, table E-VII of MIL–PRF–19500.

Subgroup	Method	Condition
C2	2036	Test condition E; (not applicable for UA and UB devices).
C5	3131	$R_{\theta JA}$ and $R_{\theta JC}$ only, as applicable (see 1.3) and in accordance with thermal impedance curves. See 4.3.3.
C6	1026	1,000 hours at V_{CB} = 10 V dc; power shall be applied to achieve T_J = +150°C minimum and a minimum of P_D = 75 percent of maximum rated P_T as defined in 1.3 n = 45, c = 0. The sample size may be increased and the test time decreased as long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours.

* 4.4.3.2 Quality levels (JAN, JANTX, and JANTXV), table E-VII of MIL–PRF–19500.

Subgroup	Method	Condition
C2	2036	Test condition E; not applicable for UA and UB devices.
C5	3131	$R_{\theta JA}$ and $R_{\theta JC}$ only, as applicable (see 1.3 and 4.3.3) and in accordance with thermal impedance curves.
C6		Not applicable.

4.4.3.3 Group C sample selection. Samples for subgroups in group C shall be chosen at random from any inspection lot containing the intended package type and lead finish procured to the same specification which is submitted to and passes table I tests herein for conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for C6 life test may be pulled prior to the application of final lead finish. Testing of a subgroup using a single device type enclosed in the intended package type shall be considered as complying with the requirements for that subgroup.

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4.4.4 Group D inspection. Conformance inspection for hardness assured JANS and JANTXV types shall include the group D tests specified in table II herein. These tests shall be performed as required in accordance with MIL–PRF–19500 and method 1019 of MIL-STD-750, for total ionizing dose or method 1017 of MIL-STD-750 for neutron fluence as applicable (see 6.2.e herein), except group D, subgroup 2 may be performed separate from other subgroups. Alternate package options may also be substituted for the testing provided there is no adverse effect to the fluence profile.

* 4.4.5 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL–PRF–19500 and as specified in table III herein; delta measurements shall be in accordance with the applicable steps of 4.6.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Collector-base time constant. This parameter may be determined by applying an rf signal voltage of 1.0 volt (rms) across the collector-base terminals and measuring the ac voltage drop (V_{eb}) with a high-impedance rf voltmeter across the emitter-base terminals. With f = 79.8 MHz used for the 1.0 volt signal, the following computation applies:

$$r'_b\ ,\ C_{c(ps)} = 2\ X\ V_{eb}\ \text{(millivolts)}$$

4.6 Delta requirements. Delta requirements shall be as follows:

Step	Inspection	MIL-STD-750		Symbol	Limit	Unit
		Method	Conditions			
1.	Collector-base cutoff current	3036	Bias condition D, V_{CB} = 60 V dc	ΔI_{CB02}	100 percent of initial value or 5 nA dc, whichever is greater.	
2.	Forward current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 100 mA dc; pulsed see 4.5.1	Δh_{FE2}	±25 percent change from initial reading.	

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* TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1 2/						
Visual and mechanical examination 3/	2071					
Solderability 3/ 4/	2026	n = 15 leads, c = 0				
Resistance to solvents 3/ 4/ 5/	1022	n = 15 devices, c = 0				
Salt atmosphere	1041	n = 6 devices, c = 0				
Temp cycling 3/ 4/	1051	Test condition C, 25 cycles, n = 22 devices, c = 0				
Hermetic seal 4/ 6/ Fine leak Gross leak	1071	n = 22 devices, c = 0				
Electrical measurements 4/		Table I, subgroup 2				
Bond strength 3/ 4/	2037	Precondition T_A = +250°C at t = 24 hours or T_A = +300°C at t = 2 hours; n = 11 wires, c = 0				
Decap internal visual (design verification) 4/	2075	n = 4 devices, c = 0				
Subgroup 2						
Thermal impedance	3131	See 4.3.3	$Z_{\theta JX}$			°C/W
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 80 V dc; pulsed (see 4.5.1)	I_{CBO1}		10	µA dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 5 V dc	I_{EBO1}		10	µA dc
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 60 V dc	I_{CBO2}		10	nA dc
Collector to emitter cutoff current	3041	Bias condition A; V_{BE} = 2.0 V dc; V_{CE} = 60 V dc	I_{CEX1}		25	nA dc
Base emitter cutoff current	3061	Bias condition D; V_{BE} = 3.0 V dc	I_{EBO2}		25	nA dc
Forward-current transfer ratio	3076	V_{CE} = 5.0 V dc; I_C = 100 µA dc	h_{FE1}	50		
Forward-current transfer ratio	3076	V_{CE} = 5.0 V dc; I_C = 100 mA dc pulsed (see 4.5.1)	h_{FE2}	100	300	
Forward-current transfer ratio	3076	V_{CE} = 5.0 V dc; I_C = 500 mA dc; pulsed (see 4.5.1)	h_{FE3}	70		
Forward-current transfer ratio	3076	V_{CE} = 5.0 V dc; I_C = 1.0 A dc; pulsed (see 4.5.1)	h_{FE4}	25		

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2 – Continued						
Collector - emitter saturated voltage	3071	I_C = 150 mA dc; I_B = 15 mA dc; pulsed (see 4.5.1)	$V_{CE(SAT)1}$		0.15	V dc
Collector - emitter saturated voltage	3071	I_C = 500 mA dc; I_B = 50 mA dc; pulsed (see 4.5.1)	$V_{CE(SAT)2}$		0.50	V dc
Collector - emitter saturated voltage	3071	I_C = 1.0 A dc; I_B = 100 mA dc; pulsed (see 4.5.1)	$V_{CE(SAT)3}$		1.0	V dc
Base - emitter saturated voltage	3066	Test condition A; I_C = 150 mA dc; I_B = 15 mA dc pulsed (see 4.5.1)	$V_{BE(SAT)1}$		0.9	V dc
Base - emitter saturated voltage	3066	Test condition A; I_C = 500 mA dc; I_B = 50 mA dc; pulsed (see 4.5.1)	$V_{BE(SAT)2}$		1.2	V dc
Subgroup 3						
High-temperature operation:		T_A = +150°C				
Collector –base cutoff current	3036	Bias condition D; V_{CB} = 60 V dc	I_{CBO3}		25	µA dc
Low-temperature operation:		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 5.0 V dc; I_C = 500 mA dc; pulsed (see 4.5.1)	h_{FE5}	30		
Subgroup 4						
Magnitude of common emitter small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 10 V dc; I_C = 50 mA dc; f = 100 MHz	$\lvert h_{fe} \rvert$	1.5	6.0	
Open circuit output capacitance	3236	V_{CB} = 10 V dc; I_E = 0; 100 kHz ≤ f ≤ 1 MHz	C_{obo}		20	pF
Input capacitance (output open-circuited)	3240	V_{EB} = 0.5 V dc; I_C = 0; 100 kHz ≤ f ≤ 1 MHz	C_{ibo}		80	pF

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4 – Continued						
Pulse response						
On-time	3251	Test condition A; I_C = 500 mA dc; I_{B1} = 50 mA dc; (see figure 19)	t_d		15	ns
Rise time	3251	Test condition A; I_C = 500 mA dc; I_{B1} = 50 mA dc; (see figure 19)	t_r		25	ns
Storage time	3251	Test condition A; I_C = 500 mA dc; I_{B1} = 50 mA dc; (see figure 20)	t_s		175	ns
Fall time	3251	Test condition A; I_C = 500 mA dc; I_{B1} = 50 mA dc; (see figure 20)	t_f		35	ns
Subgroups 5, 6, and 7						
Not applicable						

1/ For sampling plan see MIL–PRF–19500.
2/ For resubmission of failed test in subgroup 1 of table I, double the sample size of the failed test or sequence of tests. A failure in table I, subgroup 1 shall not require retest of the entire subgroup. Only the failed test shall be rerun upon submission.
3/ Separate samples may be used.
4/ Not required for JANS devices.
5/ Not required for laser marked devices.
6/ This hermetic seal test is an end-point to temp-cycling in addition to electrical measurements.

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TABLE II. Group D inspection.

Inspection 1/ 2/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1 3/						
Neutron irradiation	1017	Neutron exposure V_{CES} = 0 V				
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 80 V dc; pulsed (see 4.5.1)	I_{CBO1}		20	µA dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 5 V dc	I_{EBO1}		20	µA dc
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 60 V dc	I_{CBO2}		20	nA dc
Collector to emitter cutoff current	3041	Bias condition A; V_{BE} = 2.0 V dc; V_{CE} = 60 V	I_{CEX1}		50	nA dc
Base emitter cutoff current	3061	Bias condition D; V_{EB} = 3 V dc	I_{EBO2}		50	nA dc
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 100 µA dc	[h_{FE1}] 4/	[25]		
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 100 mA dc	[h_{FE2}] 4/	[50]	300	
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 500 mA dc; pulsed (see 4.5.1)	[h_{FE3}] 4/	[35]		
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 1 A dc; pulsed (see 4.5.1)	[h_{FE4}] 4/	[12]		
Collector-emitter saturation voltage	3071	I_C = 150 mA dc; I_B = 15 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)1}$		.18	V dc
Collector-emitter saturation voltage	3071	I_C = 500 mA dc; I_B = 50 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)2}$		.58	V dc
Collector-emitter saturation voltage	3071	Ic = 1 A dc; I_B = 100 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)3}$		1.15	V dc
Base-emitter saturation voltage	3066	Test condition A; I_C = 150 mA dc; I_B = 15 mA dc; pulsed (see 4.5.1)	$V_{BE(sat)1}$		1.1	V dc
Base-emitter saturation voltage	3066	Test condition A; I_C = 500 mA dc; I_B = 50 mA dc; pulsed (see 4.5.1)	$V_{BE(sat)2}$		1.5	V dc

See footnotes at end of table.

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TABLE II. Group D inspection - Continued.

Inspection 1/ 2/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 2						
Total dose irradiation	1019	Gamma exposure V_{CES} = 64 V Condition A				
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 80 V dc; pulsed (see 4.5.1)	I_{CBO1}		20	µA dc
Emitter to base cutoff current	3061	Bias condition D; V_{EB} = 5 V dc	I_{EBO1}		20	µA dc
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 60 V dc	I_{CBO2}		20	nA dc
Collector to emitter cutoff current	3041	Bias condition A; V_{BE} = 2.0 V dc; V_{CE} = 60 V	I_{CEX1}		50	nA dc
Base emitter cutoff current	3061	Bias condition D; V_{EB} = 3 V dc	I_{EBO2}		50	nA dc
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 100 µA dc	[h_{FE1}] 4/	[25]		
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 100 mA dc	[h_{FE2}] 4/	[50]	300	
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 500 mA dc; pulsed (see 4.5.1)	[h_{FE3}] 4/	[35]		
Forward-current transfer ratio	3076	V_{CE} = 5 V dc; I_C = 1 A dc; pulsed (see 4.5.1)	[h_{FE4}] 4/	[12]		
Collector-emitter saturation voltage	3071	I_C = 150 mA dc; I_B = 15 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)1}$		.18	V dc
Collector-emitter saturation voltage	3071	I_C = 500 mA dc; I_B = 50 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)2}$		.58	V dc
Collector-emitter saturation voltage	3071	Ic = 1 A dc; I_B = 100 mA dc; pulsed (see 4.5.1)	$V_{CE(sat)3}$		1.15	V dc
Base-emitter saturation voltage	3066	Test condition A; I_C = 150 mA dc; I_B = 15 mA dc; pulsed (see 4.5.1)	$V_{BE(sat)1}$		1.1	V dc
Base-emitter saturation voltage	3066	Test condition A; I_C = 500 mA dc; I_B = 50 mA dc; pulsed (see 4.5.1)	$V_{BE(sat)2}$		1.5	V dc

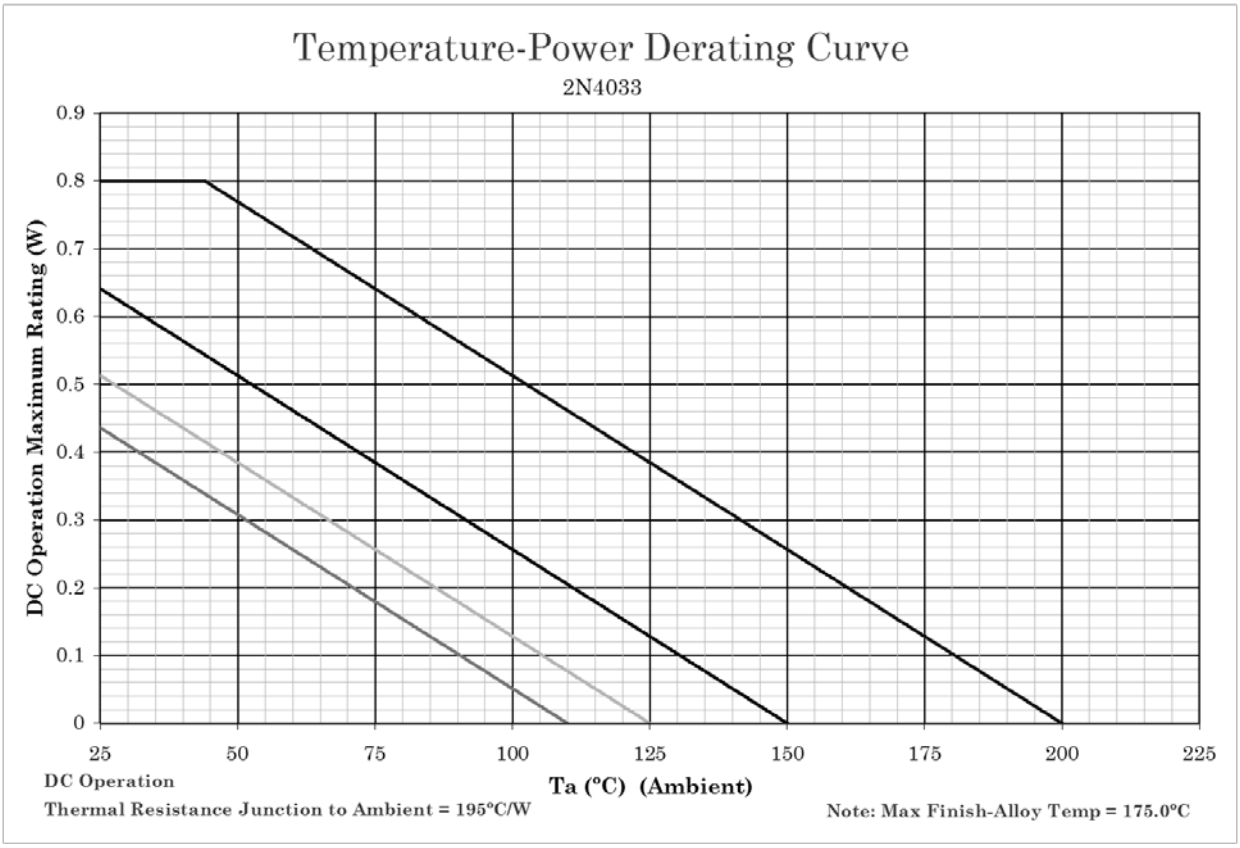
1/ Tests to be performed on all devices receiving radiation exposure.
2/ For sampling plan, see MIL–PRF–19500.
3/ See 6.2.e herein.
4/ See method 1019 of MIL-STD-750 for how to determine [h_{FE}] by first calculating the delta ($1/h_{FE}$) from the pre- and post-radiation h_{FE}. Notice the [h_{FE}] is not the same as h_{FE} and cannot be measured directly. The [h_{FE}] value can never exceed the pre-radiation minimum h_{FE} that it is based upon.

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TABLE III. Group E inspection (all quality levels) - for qualification or re-qualification only.

Inspection	MIL-STD-750		Qualification
	Method	Conditions	
Subgroup 1			45 devices c = 0
Temperature cycling (air to air)	1051	Test condition C, 500 cycles.	
Hermetic seal	1071		
Fine leak Gross leak			
Electrical measurements		See table I, subgroup 2 and 4.5.2 herein.	
Subgroup 2			45 devices c = 0
Intermittent life	1037	Intermittent operation life: V_{CB} = 10 V dc, 6,000 cycles. Adjust device current, or power, to achieve a minimum ΔT_J of +100°C.	
Electrical measurements		See table I, subgroup 2 and 4.5.2 herein.	
Subgroup 4			
Thermal resistance	3131	$R_{\theta JSP(IS)}$ can be calculated but shall be measured once in the same package with a similar die size to confirm calculations (may apply to multiple specification sheets). $R_{\theta JSP(AM)}$ need be calculated only.	15 devices, c = 0
Thermal impedance curves		See MIL–PRF–19500, table E-IX, group E, subgroup 4.	
Subgroup 5			
Not applicable			
Subgroup 6			11 devices
ESD	1020		
Subgroup 8			45 devices c = 0
Reverse stability	1033	Condition B.	

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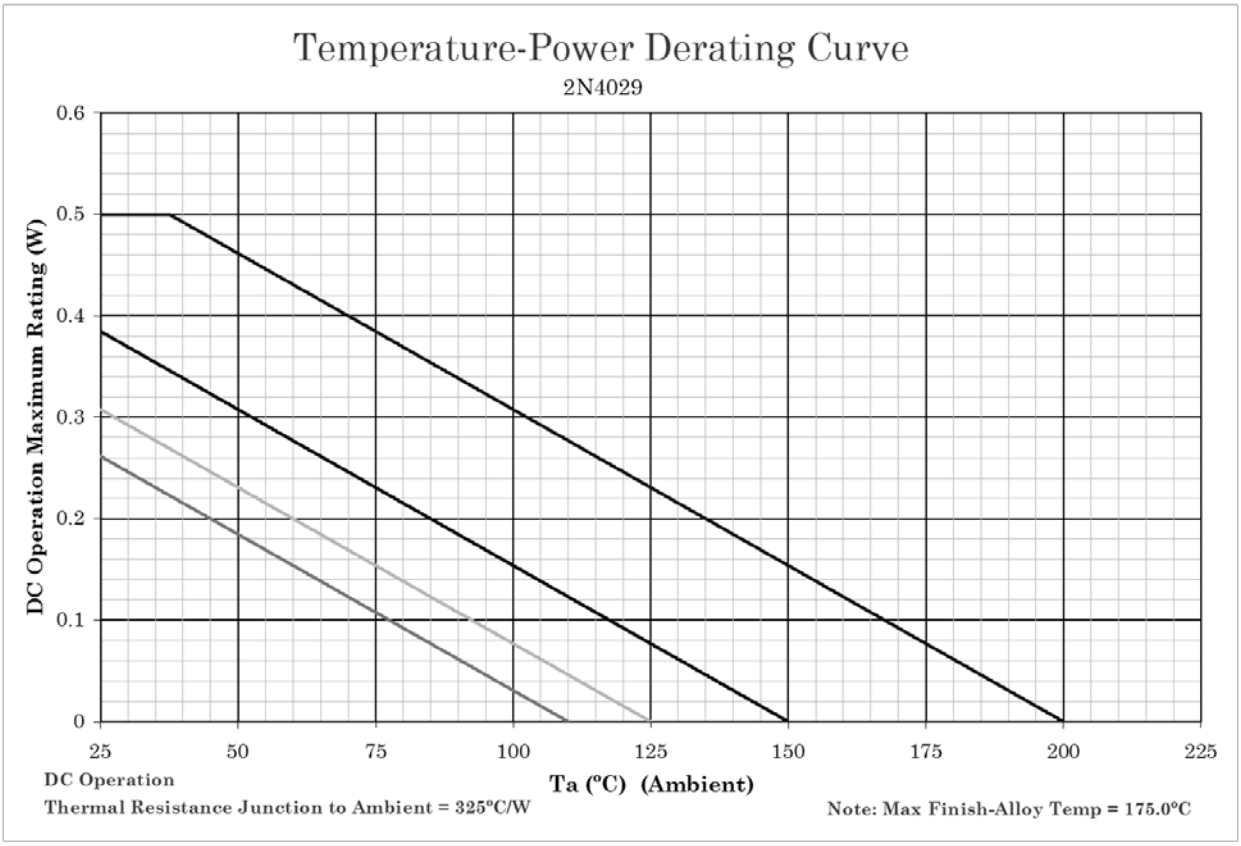


NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at $\leq T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J \leq 200°C$) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J \leq 150°C$, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at $T_J \leq$, 125°C, and 110°C to show power rating where most users want to limit T_J in their application.

FIGURE 7. Derating for 2N4033 ($R_{\theta JA}$) (TO-39).

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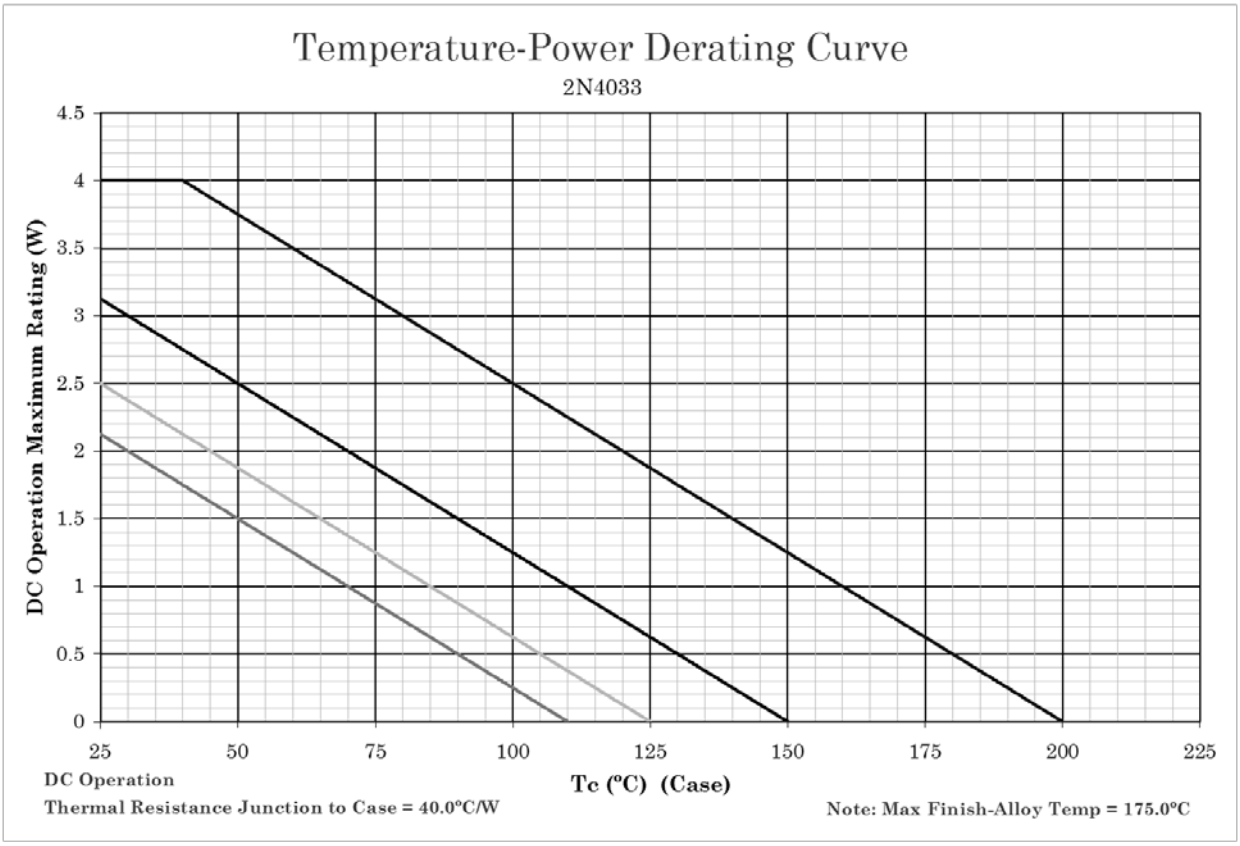


NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at $\leq T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J \leq 200°C$) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J \leq 150°C$, where the maximum temperature of electrical test is performed.
4. Derate design curves chosen at $T_J \leq$, 125°C, and 110°C to show power rating where most users want to limit T_J in their application.

FIGURE 8. Derating for 2N4029 ($R_{\theta JA}$) (TO-18), leads .125 inch (3.17 mm).

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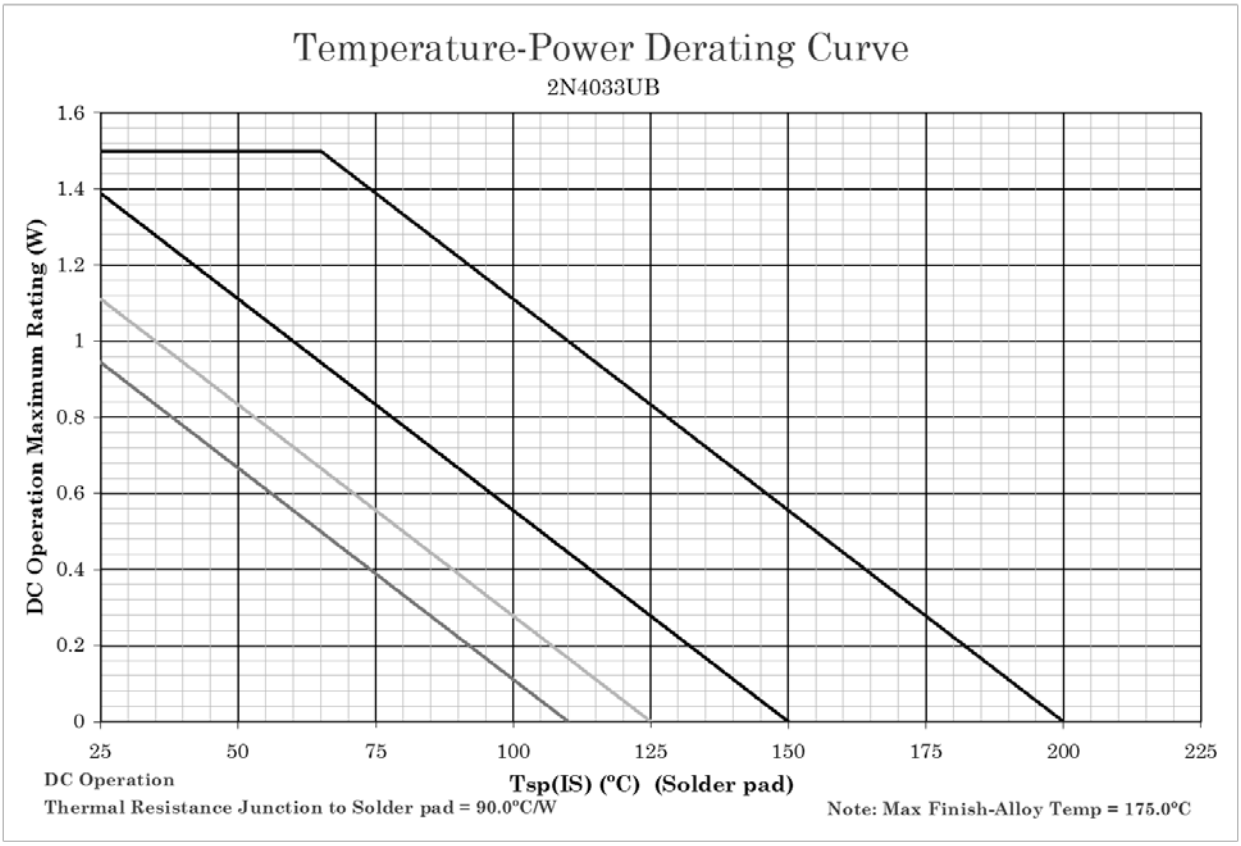


NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at $\leq T_J$ specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J \leq 200°C$) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J \leq 150°C$, where the maximum temperature of electrical test is performed.
4. Derate design curve chosen at $T_J \leq 125°C$, and 110°C to show power rating where most users want to limit T_J in their application.

FIGURE 9. Derating for 2N4033 ($R_{\theta JC}$) (TO-39).

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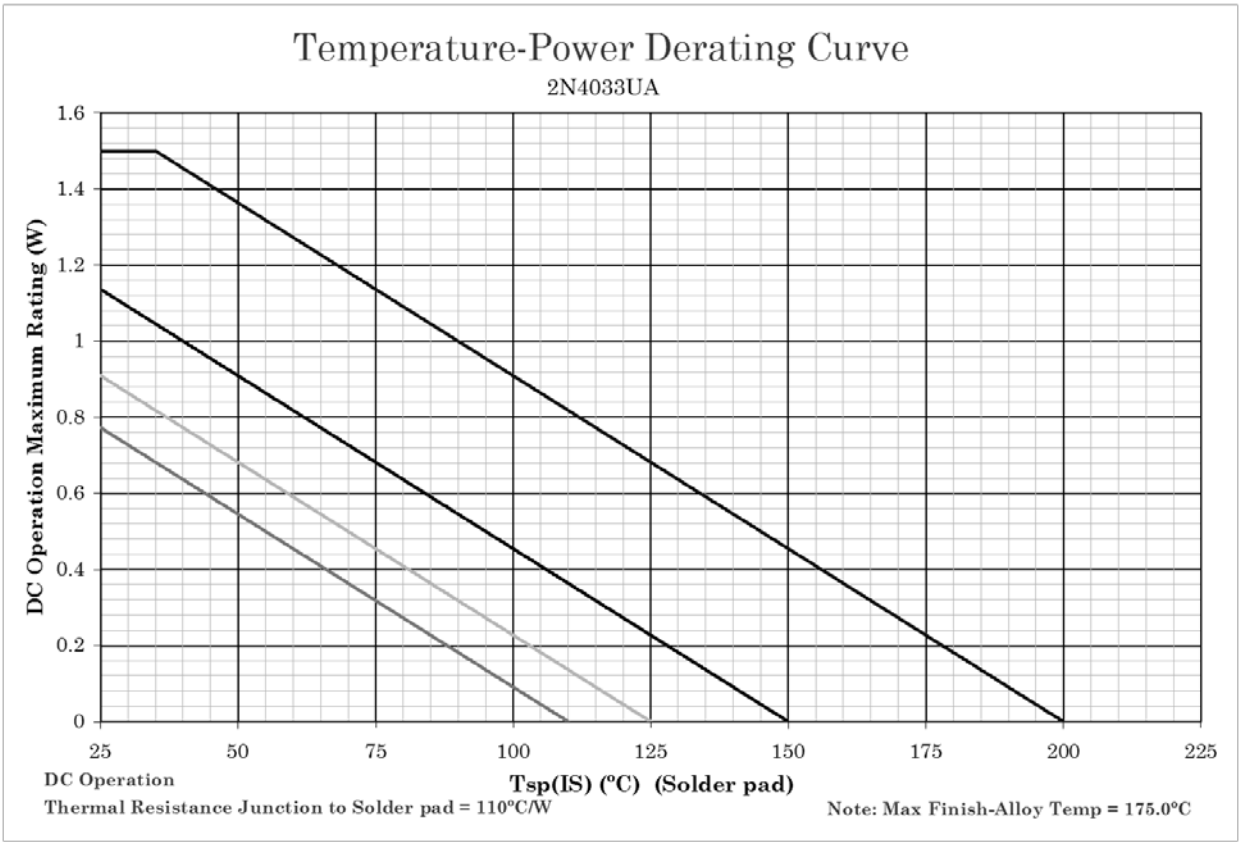


NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ T_J specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J \leq 200°C$) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J \leq 150°C$, where the maximum temperature of electrical test is performed.
4. Derate design curve chosen at $T_J \leq 125°C$, and 110°C to show power rating where most users want to limit T_J in their application.

FIGURE 10. Derating for 2N4033UB ($R_{\theta JSP(IS)}$), infinite sink 3-points.

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NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ T_J specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperature ($T_J \leq 200°C$) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at $T_J \leq 150°C$, where the maximum temperature of electrical test is performed.
4. Derate design curve chosen at $T_J \leq 125°C$, and 110°C to show power rating where most users want to limit T_J in their application.

FIGURE 11. Derating for 2N4033UA ($R_{\theta JSP(IS)}$).

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Maximum Thermal Impedance

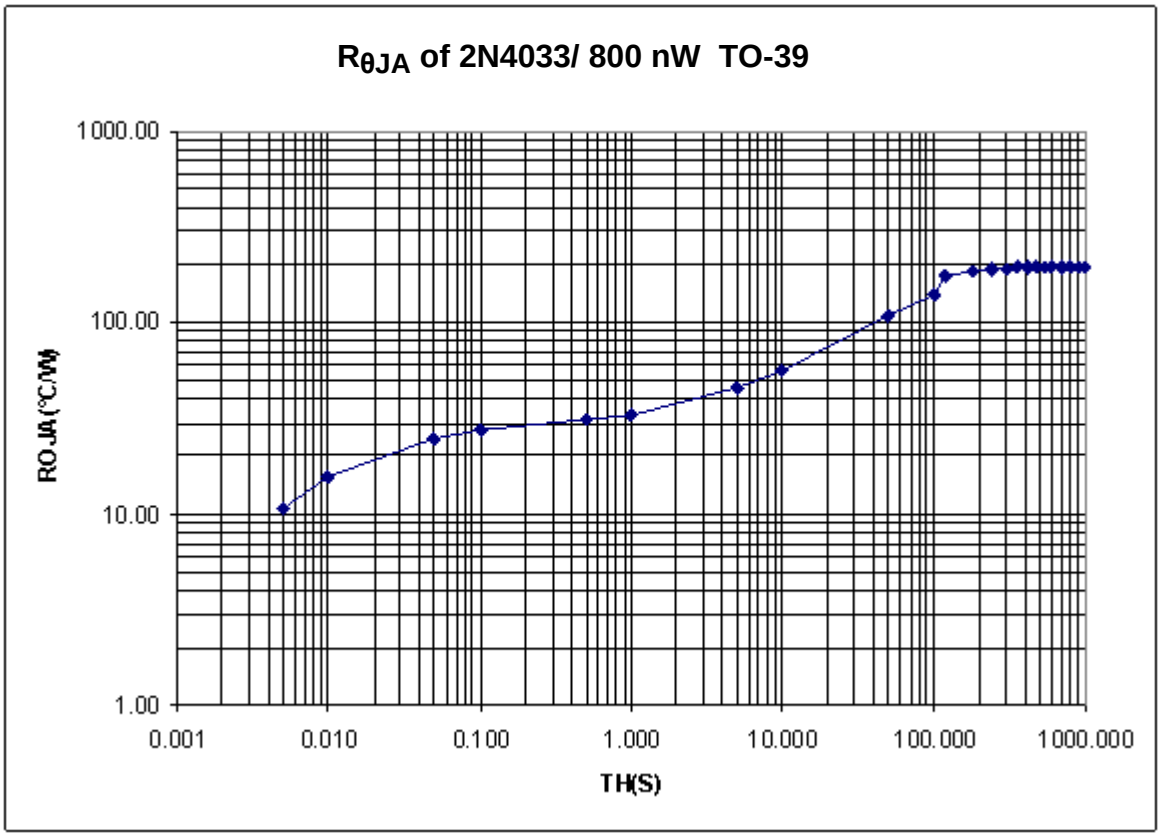


FIGURE 12. Thermal impedance graph ($R_{\theta JA}$) for 2N4033 (TO-39).

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Maximum Thermal Impedance

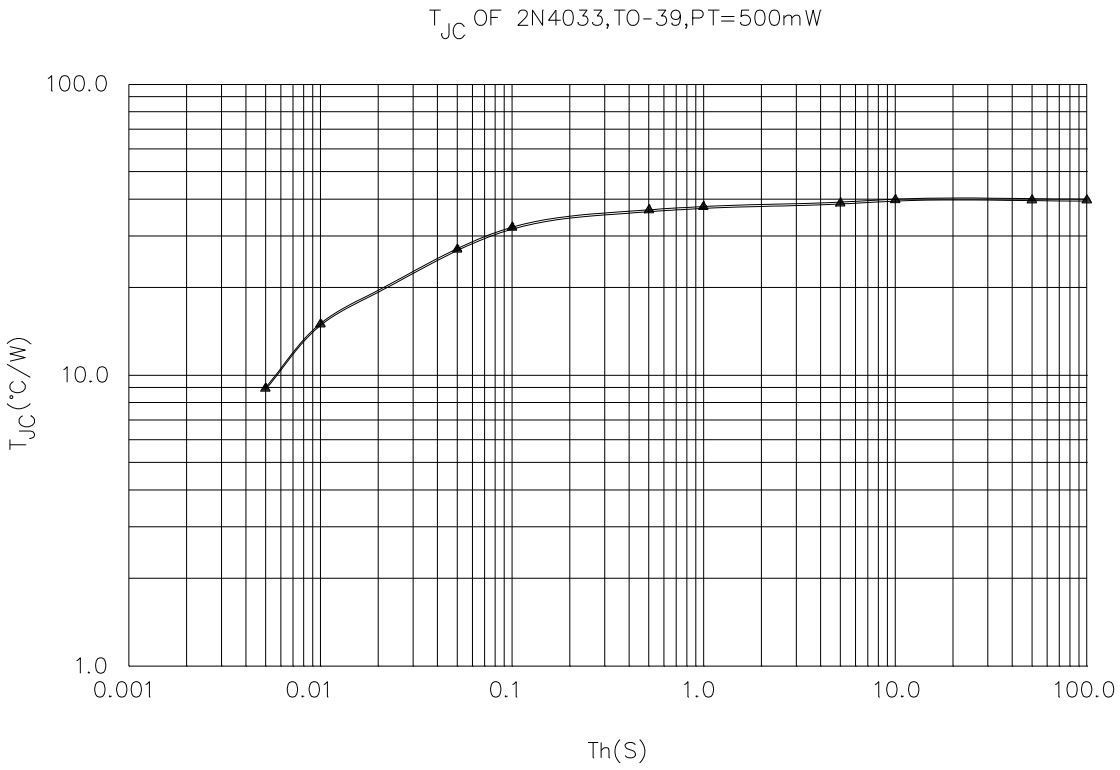


FIGURE 13. Thermal impedance graph ($R_{\theta JC}$) for 2N4033 (TO-39).

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Maximum Thermal Impedance

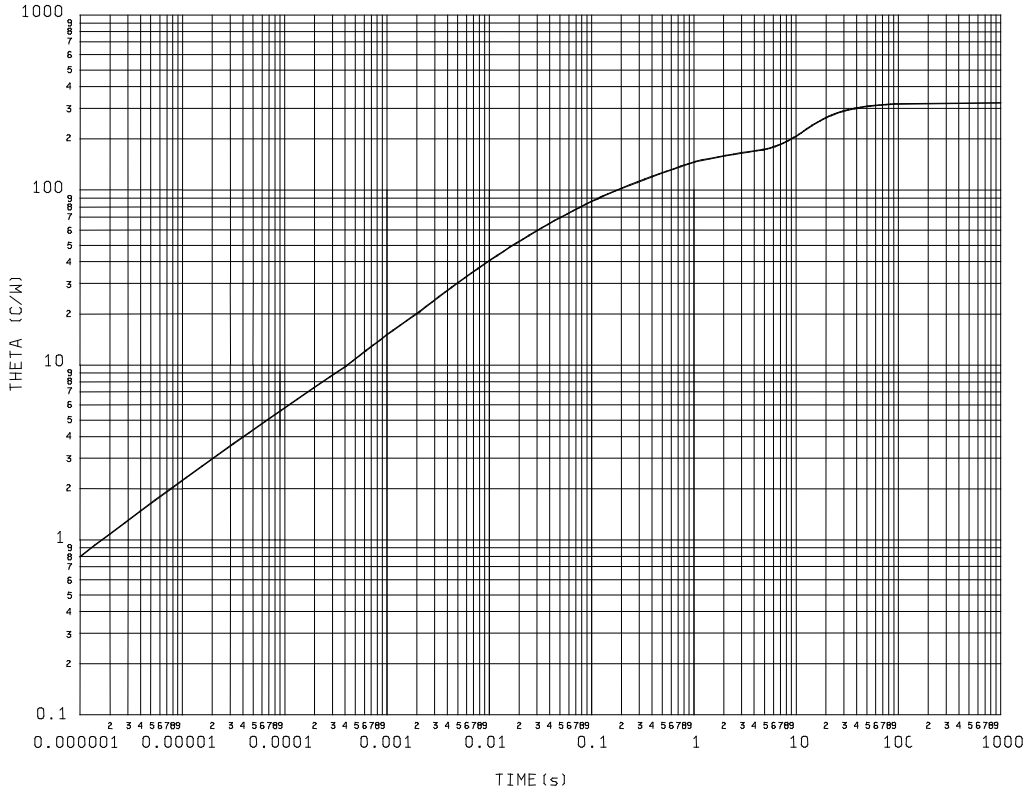
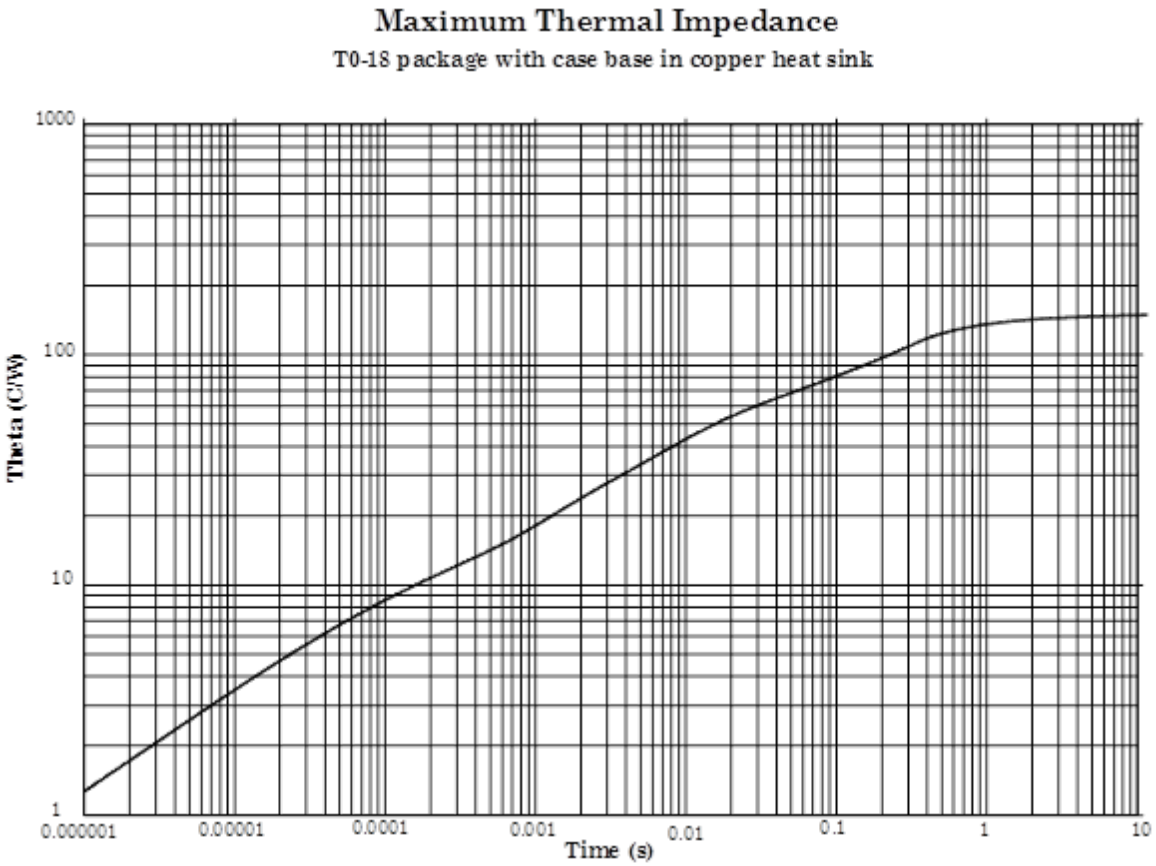


FIGURE 14. Thermal impedance graph ($R_{\theta JA}$) for 2N4029 (TO-18).

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$R_{\theta JC}$ = 150°C/W

FIGURE 15. Thermal impedance graph ($R_{\theta JC}$) for 2N4029 (TO-18).

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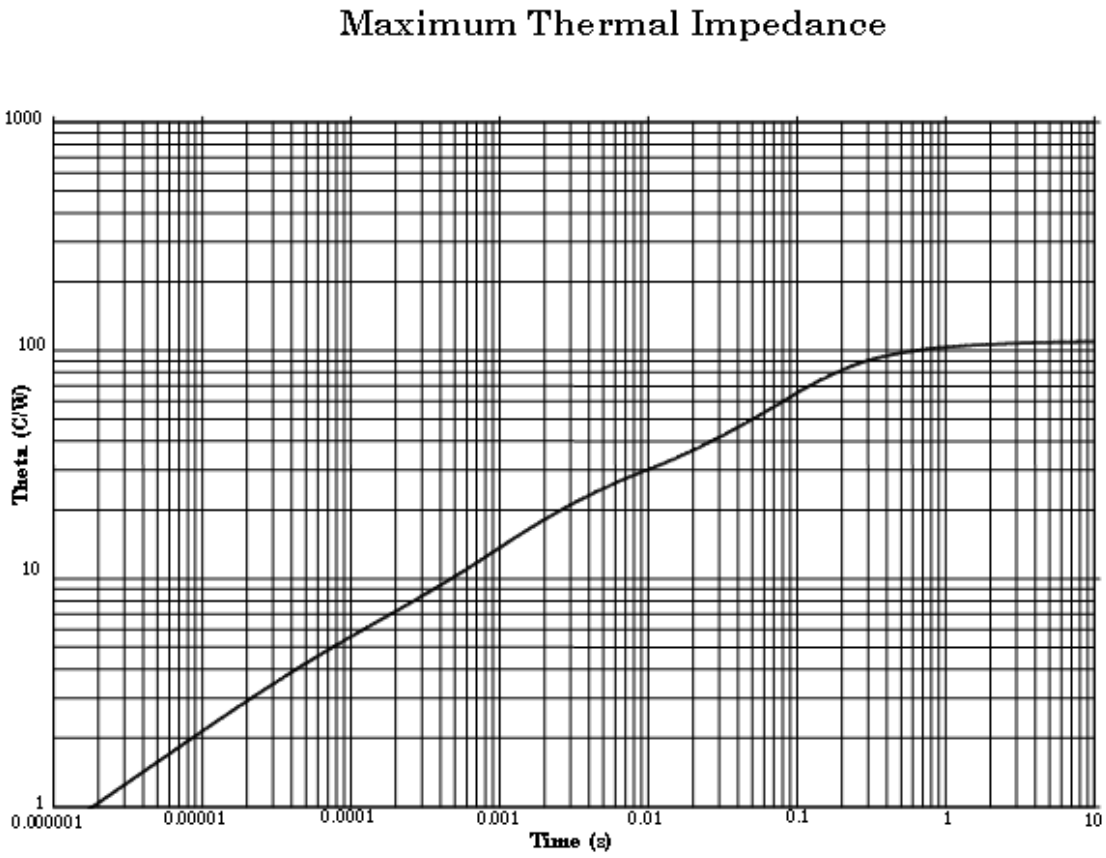
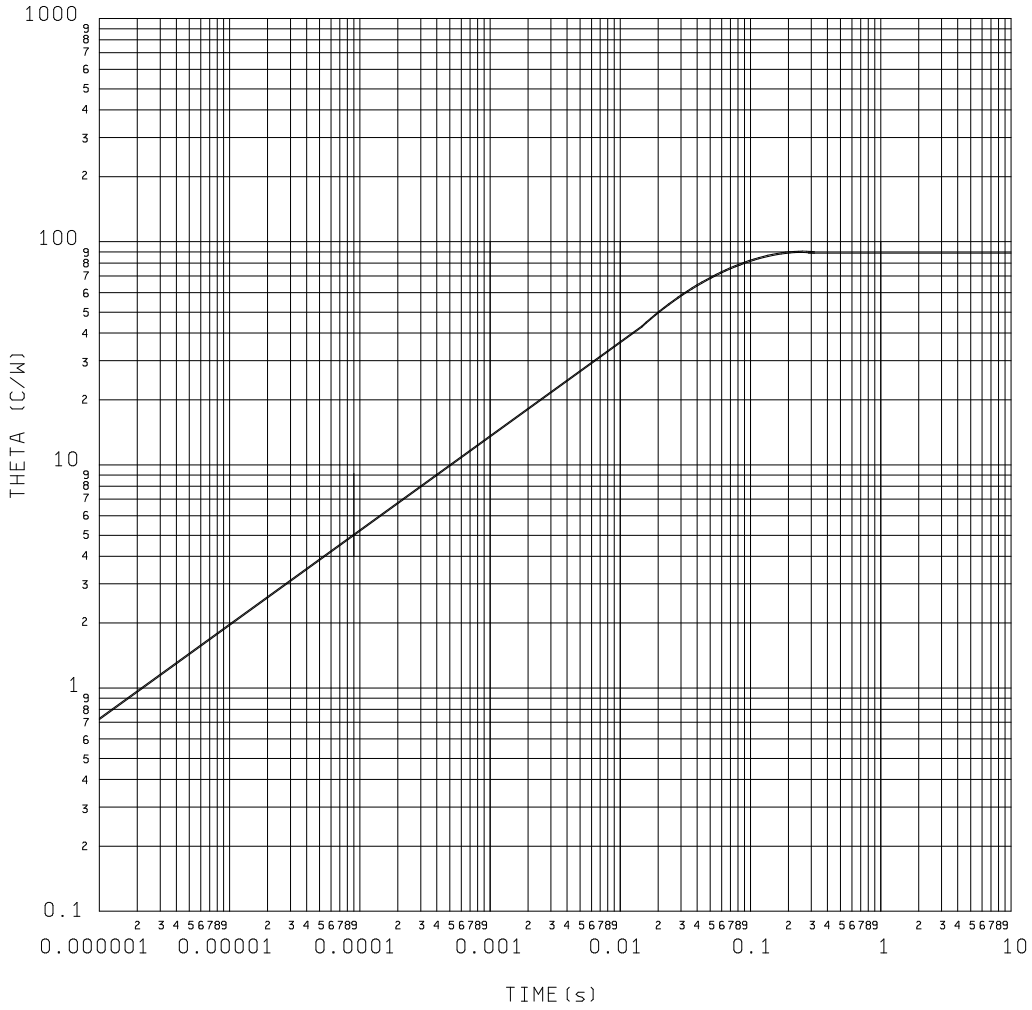


FIGURE 16. Thermal impedance graph ($R_{\theta JSP(IS)}$) for 2N4033 (UA).

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Check the source to verify that this is the current version before use.

Maximum Thermal Impedance

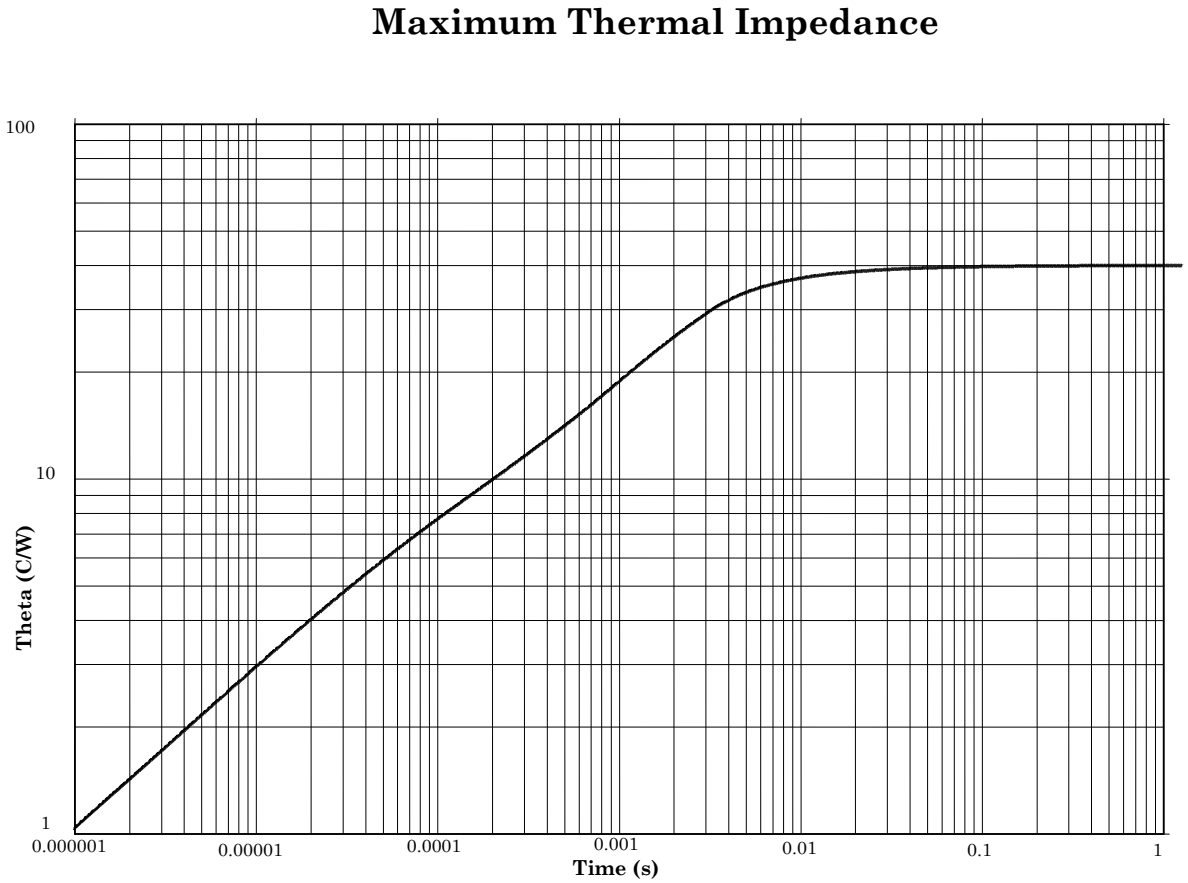


Ceramic UB package soldered to PCB 3 points solder pad (infinite sink to PCB).

$R_{\theta JSP(IS)}$ = 90°C/W

FIGURE 17. Thermal impedance graph ($R_{\theta JSP(IS)}$) for 2N4029 (UB).

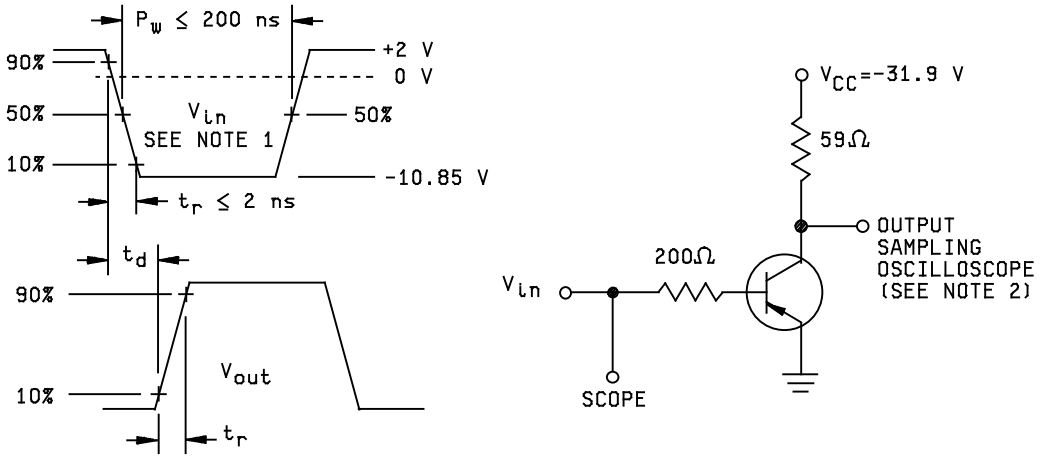
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2N4033UA 4 point solder pad (adhesive mount to PCB), $R_{\theta JSP(AM)}$ = 40°C/W

FIGURE 18. Thermal impedance graph $R_{\theta JSP(AM)}$ for 2N4033UA.

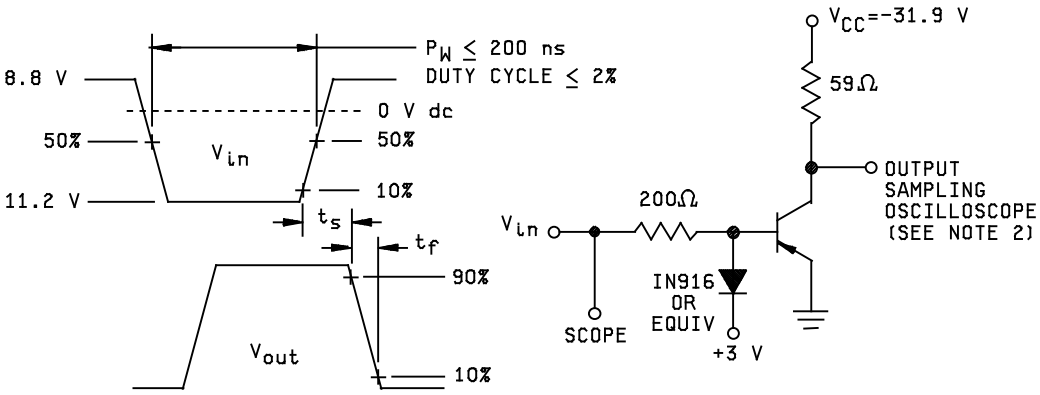
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NOTES:

1. The rise time (t_r) of the applied pulse shall be ≤ 2.0 ns, duty cycle ≤ 2 percent, and the generator source Z shall be 50 Ω.
2. Sampling oscilloscope: Z_{IN} ≥ 100 kΩ; C_{in} ≤ 12 pF, rise time(t_r) ≤ 5 ns.

FIGURE 19. Delay and rise time, test circuit.



NOTES:

1. The rise time (t_r) of the applied pulse shall be ≤ 20 ns, duty cycle ≤ 2 percent, and the generator source impedance shall be 50Ω.
2. Sampling oscilloscope: Z_{IN} ≥ 100 kΩ; C_{in} ≤ 12 pF, rise time (t_r) ≤ 5 ns.

FIGURE 20. Storage and fall time, test circuit.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL–PRF–19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

* 6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. The complete PIN (see 1.5 and 6.5) .

e. For acquisition of RHA designated devices, table II, subgroup 1 testing of group D herein is optional. If subgroup 1 is desired, it must be specified in the contract.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: /VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil . An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

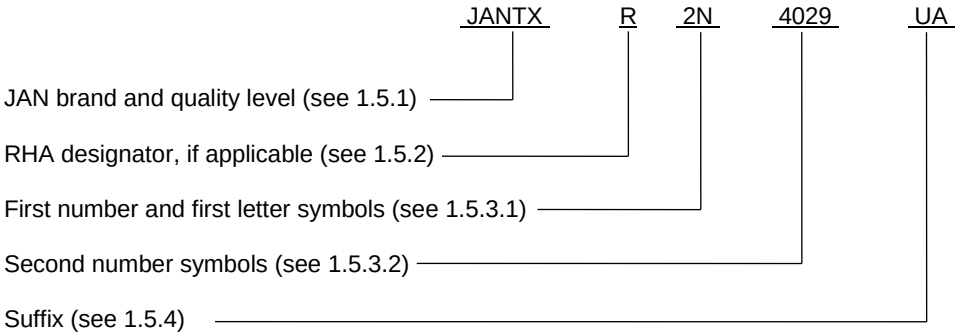
* 6.4. Suppliers of JANHC and JANKC die. The qualified JANHC and JANKC suppliers with the applicable letter version (example JANHCA2N4033) will be identified on the QPDSIS.

Die ordering information		
PIN	Manufacturer	
	34156	43611
2N4033	JANHCA2N4033 JANKCA2N4033	JANHCB2N4033 JANKCB2N4033

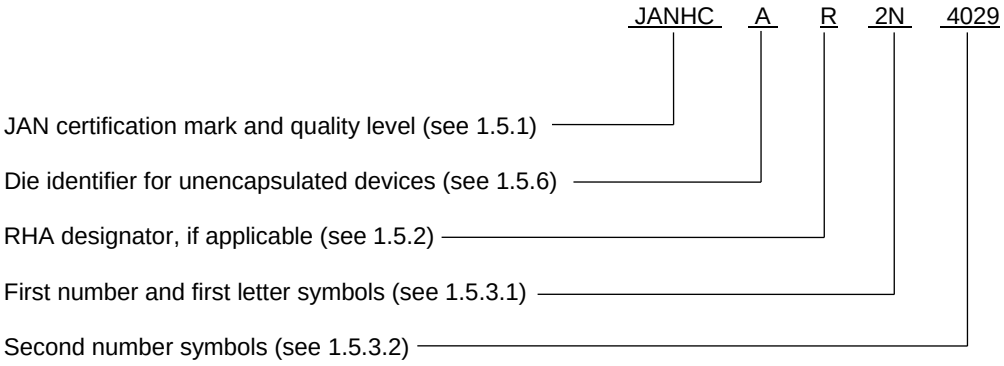
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* 6.5 PIN construction example.

* 6.5.1 PIN construction example for encapsulated devices. The PINs for encapsulated devices are construction using the following form.



* 6.5.2 Unencapsulated devices. The PINs for un-encapsulated devices are constructed using the following form.



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* 6.6 List of PINs.

* 6.6.1 List of PINs for encapsulated devices. The following is a list of possible PINs for encapsulated devices available on this specification sheet.

PINs for devices of the base quality level	PINs for devices of the "TX" quality level	PINs for devices of the "TXV" quality level (1)	PINs for devices of the "S" quality level (1)
JAN2N4029	JANTX2N4029	JANTXV#2N4029	JANS#2N4029
JAN2N4029UA	JANTX2N4029UA	JANTXV#2N4029UA	JANS#2N4029UA
JAN2N4029UB	JANTX2N4029UB	JANTXV#2N4029UB	JANS#2N4029UB
JAN2N4033	JANTX2N4033	JANTXV#2N4033	JANS#2N4033
JAN2N4033UA	JANTX2N4033UA	JANTXV#2N4033UA	JANS#2N4033UA
JAN2N4033UB	JANTX2N4033UB	JANTXV#2N4033UB	JANS#2N4033UB

(1) The number sign (#) represent one of eight RHA designators available (M, D, P, L, R, F, G, or H) if desired. Remove for no RHA.

* 6.6.2 List of PINs for unencapsulated devices. The following is a list of possible PINs for unencapsulated devices available on this specification sheet.

JANHCA#2N4029	JANHCA#2N4033
JANKCA#2N4029	JANKCA#2N4033
JANHCB#2N4029	JANHCB#2N4033
JANKCB#2N4029	JANKCB#2N4033

(1) The number sign (#) represent one of eight RHA designators available (M, D, P, L, R, F, G, or H) if desired. Remove for no RHA.

6.7 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5961-2015-051)

Review activities:
Army - AV, MI
Air Force - 19, 71, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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